

Mutual Exclusion – Distributed Algorithm

```
#include<stdio.h>
#include<mpi.h>
#include<string.h>

int main()
{

MPI_Init(NULL, NULL);

int rank, size, indv_ts, all_ts[3], i;
int min, minpid, count=0;

char msg[10];
MPI_Status s;

MPI_Comm_rank(MPI_COMM_WORLD, &rank);
MPI_Comm_size(MPI_COMM_WORLD, &size);

printf("\n Enter timestamp for process%d:", rank);
scanf("%d",&indv_ts);

// send its timestamp
for(i=0;i<size;i++)
{
MPI_Send(&indv_ts, 1, MPI_INT, i, 111, MPI_COMM_WORLD);
}

// receive all timestamp
for(i=0;i<size;i++)
{
MPI_Recv(&all_ts[i], 1, MPI_INT, i, 111, MPI_COMM_WORLD, &s);
}

min = all_ts[0];
for(i=1;i<size;i++)
{
if(all_ts[i] < min)
{
min = all_ts[i];
minpid = i;
}
}

strcpy(msg,"OK");
MPI_Send(msg, sizeof(msg), MPI_CHAR, minpid, 222, MPI_COMM_WORLD);
```

```

if(rank == minpid)
{
for(i=0;i<(size-1);i++)
{
MPI_Recv(msg, sizeof(msg), MPI_CHAR, MPI_ANY_SOURCE, 222, MPI_COMM_WORLD,
&s);
if(strcmp(msg, "OK") == 0)
{
count++;
}
}
}

if(count == 2)
{
printf("\n process%d got permission to access shared data", rank);
}
}

MPI_Finalize();
return 0;
}

```

Output

```

mpicc mutual_exclusion.c -o me
mpirun -np 3 me

```

```

Enter timestamp for process0: Enter timestamp for process1:
Enter timestamp for process2:5
3
4

```

```

process1 got permission to access shared data

```